

25.7 Summary and Notes

25.7.1 Summary

Idea.

1. $E(Y \mid \text{do}(X = x))$, defined by a structural causal model (Definition 25.1.1), differs from $E(Y \mid X = x)$ under confounding, at every sample size (Proposition 25.1.1).
2. Potential outcomes (Definition 25.2.1): the naive comparison is the effect on the treated plus a selection bias (Proposition 25.2.1).
3. Under complete randomization the difference in means is unbiased, with Neyman's variance; the estimate $s_1^2/n_1 + s_0^2/n_0$ is conservative and equals the HC2 sandwich (Theorem 25.2.3).
4. An intervention deletes one factor of the Markov factorization (Proposition 25.3.1); linear effects are sums over paths (Proposition 25.3.2); d-separation reads conditional independence off the graph (Proposition 25.3.3, Theorem 25.3.4).
5. Confounding is an open back-door path, and in linear models its bias is a sum over common ancestors (Proposition 25.3.5).
6. An adjustment set identifies the effect; in linear models the coefficient of X is then the total effect (Theorem 25.4.1). Causes of Y improve precision, pure causes of X worsen it (Proposition 25.4.2).
7. Mediators, common effects and pre-treatment colliders are bad controls (Theorem 25.5.1).
8. In experiments, analysis of covariance can lose to no adjustment, and the interacted regression is asymptotically optimal (Lemma 25.6.1, Theorem 25.6.2).

25.7.2 Notes and sources

Structural models and graphs. Wright (1921, 1934) introduced path analysis (Proposition 25.3.2), and Haavelmo (1943) read structural equations as autonomous mechanisms that interventions change one at a time. The intervention as an explicit operation and the truncated factorization (Proposition 25.3.1) are from Pearl (1995), and in parallel the manipulation theorem of Spirtes, Glymour and Scheines (2000; first edition 1993); Pearl (2009) is the systematic account. d-separation is due to Pearl (1988), with proofs by Geiger, Verma and Pearl (1990) and Lauritzen, Dawid, Larsen and Leimer (1990); Spirtes, Glymour and Scheines (2000) develop faithfulness.

Potential outcomes. Definition 25.2.1 and Eq. 25.2.3 are Neyman's (1923), in a paper on field trials translated in 1990. Rubin (1974) extended the framework to observational studies, Holland (1986) named the fundamental problem of causal inference, and Imbens and Rubin (2015) set out the finite-population view. Finite-population central limit theorems go back to Hájek (1960); Li and Ding (2017) give forms suited to causal inference. Samii and Aronow (2012) show that Neyman's estimate is the HC2 sandwich (Exercise (B2)).

Adjustment and bad controls. The back-door and front-door criteria are from Pearl (1995); Rosenbaum and Rubin (1983) introduced strong ignorability and the propensity score; Shpitser, VanderWeele and Robins (2010) give a necessary and sufficient adjustment criterion; and Henckel, Perković and Maathuis (2022) find the most efficient adjustment set in linear

models. Exercise (B1) is Angrist (1998)’s observation, and the “Table 2 fallacy” is named by Westreich and Greenland (2013). Angrist and Pischke (2009) popularized the term bad control. Berkson (1946) described selection on a common effect, Greenland, Pearl and Robins (1999) showed that pre-treatment colliders can bias adjustment, Ding and Miratrix (2015) quantify M-bias and butterfly bias, Elwert and Winship (2014) survey collider bias, and Pearl (2010) analyses bias amplification.

Adjustment in experiments. Covariance adjustment of experiments is Fisher’s (1935). The design-based critique is Freedman (2008a, 2008b), and the interacted estimator and its sandwich standard errors are Lin (2013). The route through the exact variance of fixed-coefficient estimators (Lemma 25.6.1), which reduces the comparison to completing a square, is this book’s.

25.7.3 References

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